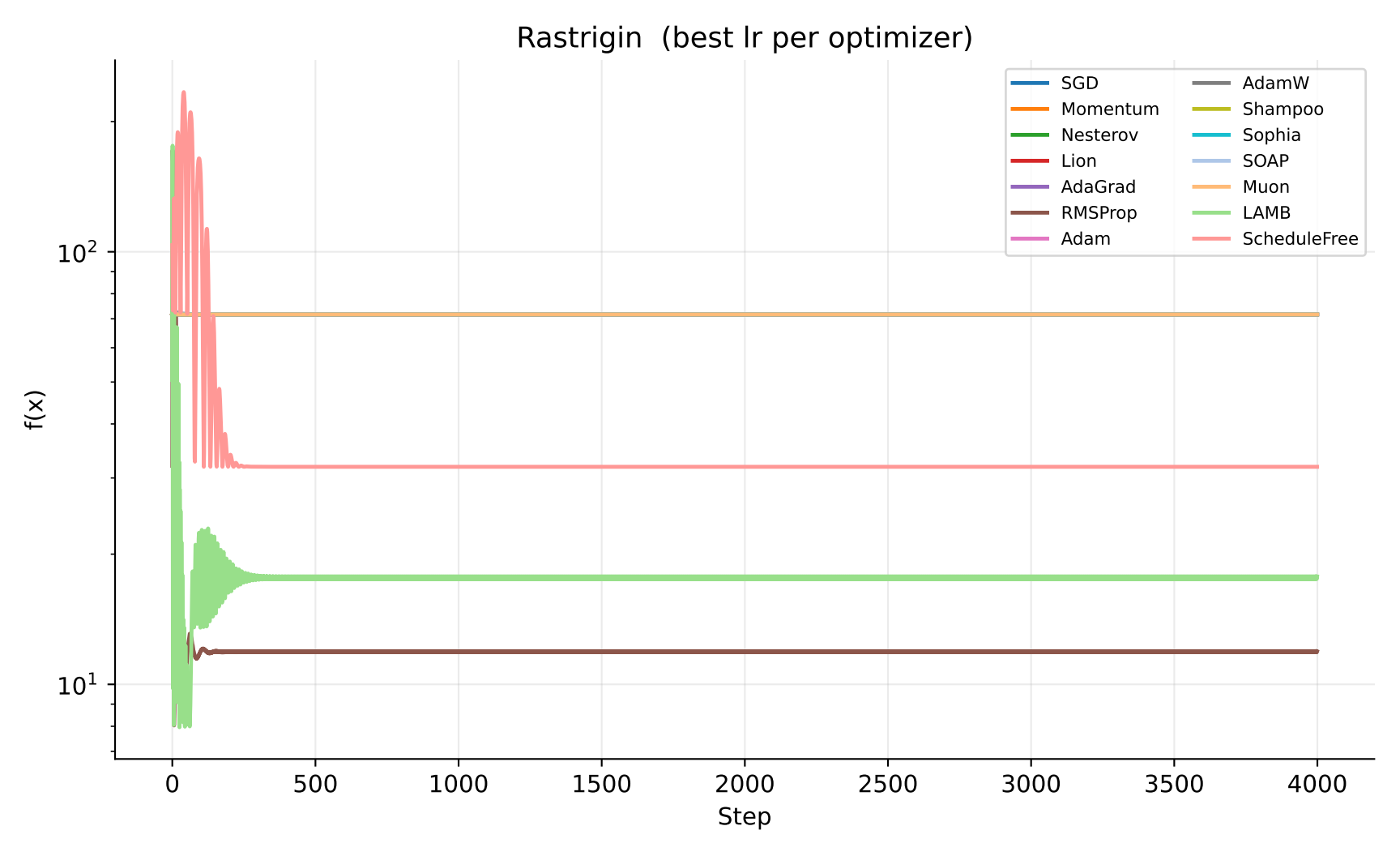

Rastrigin (best lr per optimizer)
SGD
Momentum
Nesterov
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
ScheduleFree
10^2
10^1
f(x)
0
500
1000
1500
2000
2500
3000
3500
4000
Step